Prediction of the Bonding States of Cysteines Using the Support Vector Machines Based on Multiple Feature Vectors and Cysteine State Sequences

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Abstract (1/2)

- The support vector machine (SVM) method is used to predict the bonding states of cysteines. Besides using local descriptors such as the local sequences, we include global information, such as amino acid compositions and the patterns of the states of cysteines (bonded or nonbonded), or cysteine state sequences, of the proteins. We found that SVM based on local sequences or global amino acid compositions yielded similar prediction accuracies for the data set comprising 4136 cysteine-containing segments extracted from 969 nonhomologous proteins.

Abstract (2/2)

- However, the SVM method based on multiple feature vectors (combining local sequences and global amino acid compositions) significantly improves the prediction accuracy, from 80% to 86%. If coupled with cysteine state sequences, SVM based on multiple feature vectors yields 90% in overall prediction accuracy and a 0.77 Matthews correlation coefficient, around 10% and 22% higher than the corresponding values obtained by SVM based on local sequence information.

Feature vectors (1/2)

- Sequence input vector: $x_s(w)$
  - A 20 binary-element vector each residue with window size w.

- Composition input vector: $x_c(w)$
  - A 20 element vector of compositional percentage of amino acid by $x_a = n_a/w$.

Feature vectors (2/2)

- Sequence input vector: $x_r(w)$
  - Graph shape index: complexity, branching, and symmetry of the group.
  - Polarizability: molar refractivity.
  - Volume: van der Waals volume of the side-chain to that of CH_2 group.
  - Hydrophobicity: *log P (amino acid) - log P (glycine)*
  - Isoelectric point.

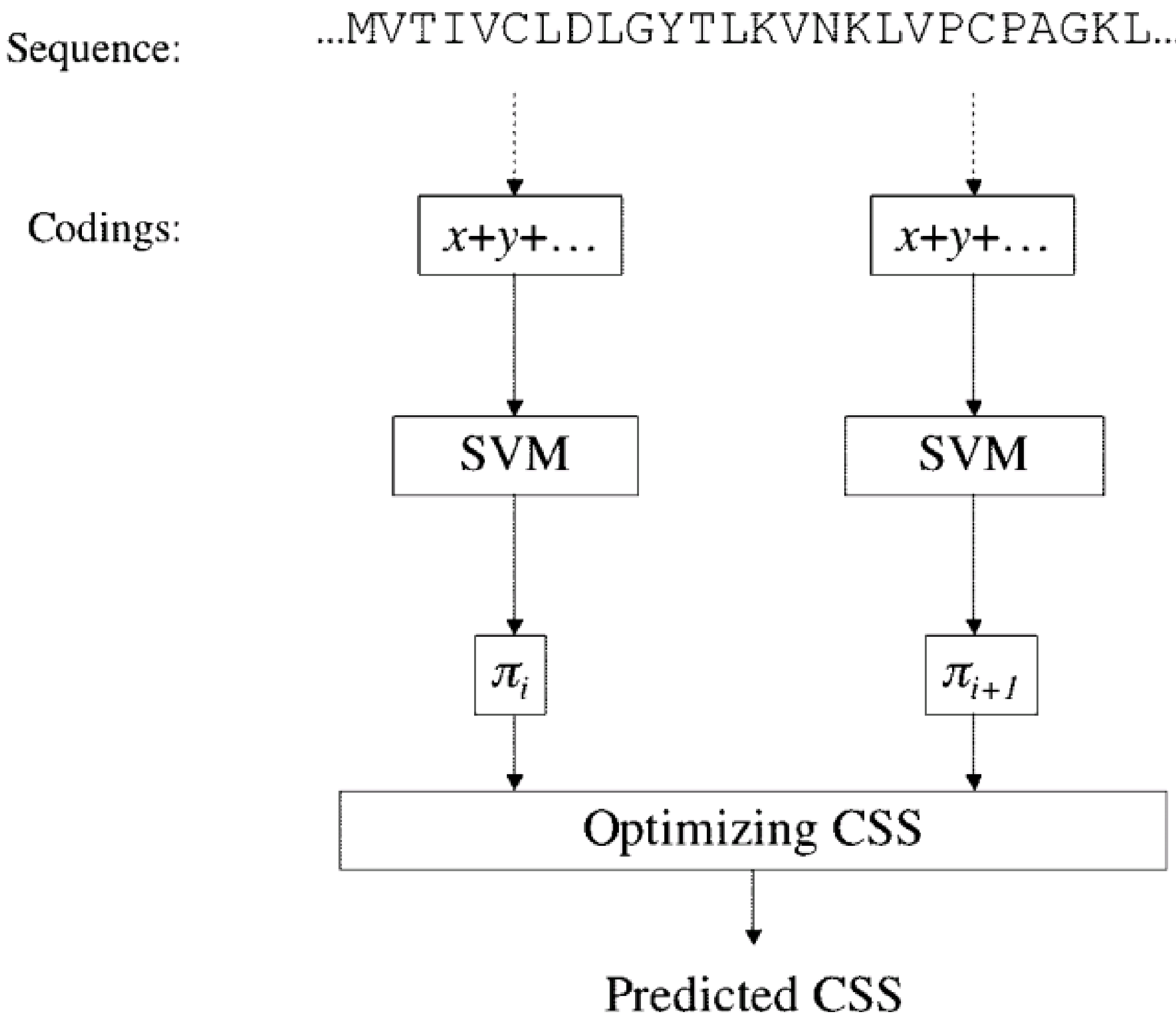
Data Sets

- 969 nonhomologous proteins from PDB
  - 4136 cysteine-containing segments
    - 1446 are in the disulfide-bonded states
    - 2690 are in the non-disulfide-bonded states

Cysteine State Sequence

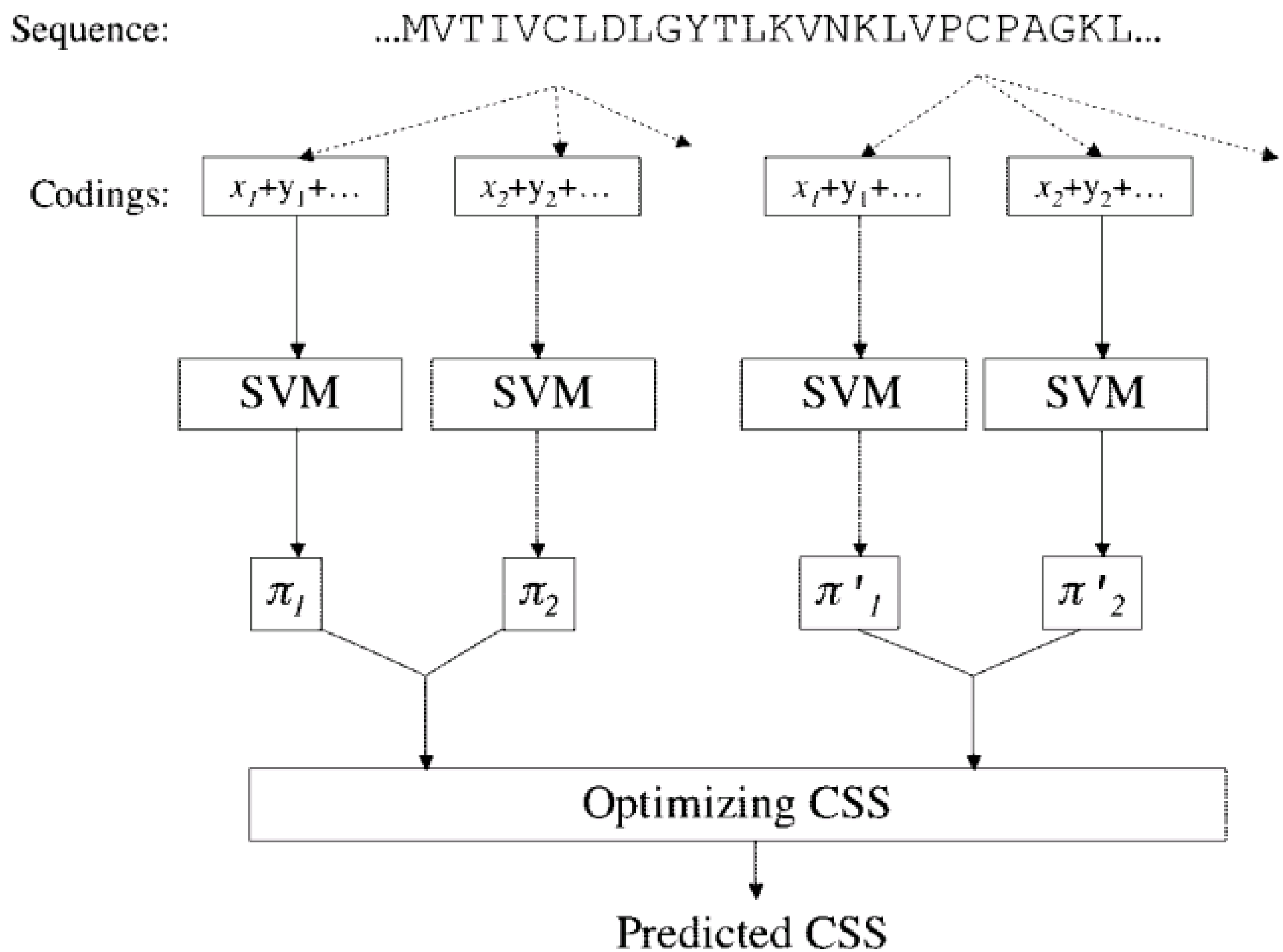
- Each Cysteine:
  - O: bonding state (Oxidized)
  - R: nonbonding state (Reduced)
- Excluding the interchain disulfide bridges
  - There are 2^{n-1} combinations.
  - For example: 3 cysteine, there are 4 cysteine state sequence
    - (OOR), (ORO), (ROO), and (RRR)
- Coding with S(start) and F(finish).

Method (1/3)



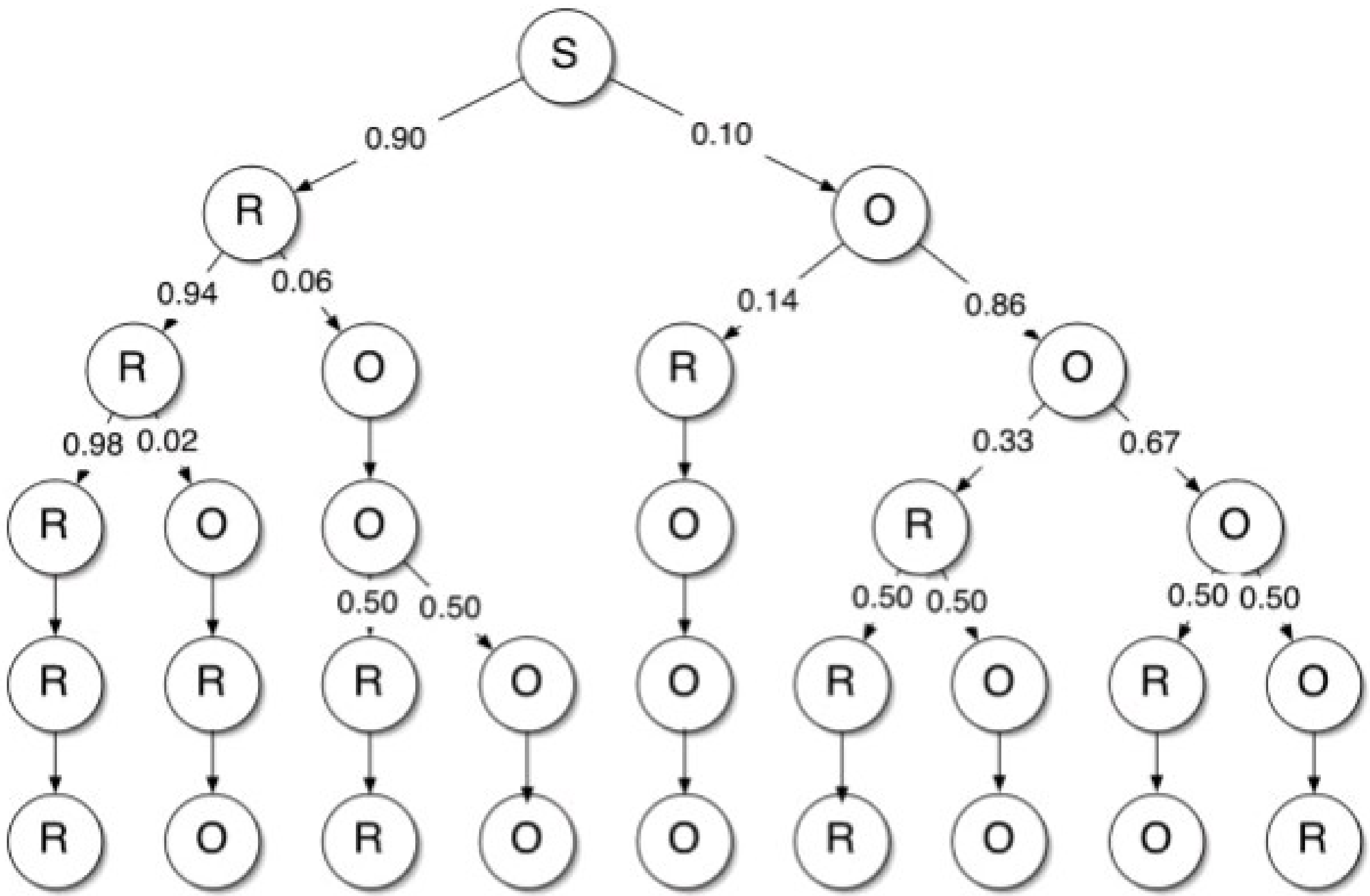
Sequence:
...MVTIVCLDLGYTLKVNKLVPCPAGKL...
Codings:
x+y+...
x+y+...
SVM
SVM
π_i
π_{i+1}
Optimizing CSS
Predicted CSS

Method (2/3)



Sequence:
...MVTIVCLDLGYTLKVNKLVPCPAGKL...
Codings:
$x_1+y_1+\ldots$
$x_2+y_2+\ldots$
$x_1+y_1+\ldots$
$x_2+y_2+\ldots$
SVM
SVM
SVM
SVM
π_1
π_2
π'_1
π'_2
Optimizing CSS
Predicted CSS

Method (3/3)


S
0.90
0.10
R
O
0.94
0.06
0.14
0.86
R
O
R
O
0.98
0.02
0.33
0.67
R
O
O
O
R
O
0.50
0.50
0.50
0.50
0.50
0.50
R
R
R
O
O
R
O
R
O
R
O
R
O
O
R
O
O
O
R

Performance Criteria

- $Q_2 = \frac{\textit{Correct predict bonding state}}{\textit{Total bonding state}}$
- $spec = \frac{TP_x}{TP_x + FP_x}, \quad x = SS \text{ or } SH,$
- $sens = \frac{TP_x}{TP_x + FN_x}, \quad x = SS \text{ or } SH,$
- $\mathrm{MCC} = \frac{TP_x TN_x - FP_x FN_x}{\sqrt{(TP_x + FN_x)(TP_x + FP_x)(TN_x + FP_x)(TN_x + FN_x)}},$

$$x = SS \text{ or } SH,$$

Result (1/2)

TABLE I. Predictive Performance of SVM Based on Sequence Inputs of Different Window Lengths, $\{x_s(w)\}$

Window length	Q_2	MCC	*SS* *spec*	*SS* *sens*	*SH* *spec*	*SH* *sens*
5	0.75	0.46	0.69	0.63	0.78	0.82
9	0.79	0.54	0.71	0.69	0.84	0.84
13	0.80	0.56	0.71	0.71	0.85	0.84
15	0.81	0.58	0.71	0.73	0.87	0.84
17	0.80	0.56	0.71	0.72	0.86	0.84
21	0.80	0.55	0.69	0.72	0.86	0.83

TABLE II. Predictive Performances of SVMs Based on Composition Inputs of Different Window Lengths, $\{x_c(w)\}$

Window length	Q_2	MCC	*SS* *spec*	*SS* *sens*	*SH* *spec*	*SH* *sens*
9	0.67	0.22	0.29	0.59	0.88	0.69
15	0.70	0.31	0.41	0.62	0.87	0.73
25	0.73	0.38	0.46	0.67	0.89	0.75
35	0.75	0.42	0.51	0.69	0.89	0.76
Full length	0.80	0.55	0.65	0.75	0.88	0.82

Result (2/2)

TABLE III. Comparison of Predictive Performances of SVM Based on Single and Multiple Feature Vectors

			SS		*SH*	
Mehods	Q_2	MCC	*spec*	*sens*	*spec*	*sens*
$\{x_s(15)\}$	0.81	0.58	0.71	0.73	0.87	0.84
$\{x_{c0}\}$	0.80	0.55	0.65	0.75	0.88	0.82
$\{x_s(7) + x_{c0}\}$	0.86	0.69	0.79	0.75	0.89	0.89

TABLE IV. Comparison of Predictive Performances of SVMs Coupled with CSS

			SS		*SH*	
Methods	Q_2	MCC	*spec*	*sens*	*spec*	*sens*
$\{x_s(15)\}$ + CSS	0.89	0.74	0.91	0.74	0.87	0.97
$\{x_s(7) + x_{c0}\}$ + CSS	0.88	0.71	0.88	0.74	0.87	0.95
$\{x_s(7) + x_r(7)\}$ + CSS	0.88	0.73	0.89	0.74	0.87	0.95
Multiple SVM + CSS[a]	0.90	0.77	0.91	0.77	0.89	0.97

[a]The multiple SVM classifiers are $\{x_s(15)\}$, $\{x_s(7) + x_{c0}\}$, and $\{x_s(7) + x_r(7)\}$.